



PRODUCT REQUIREMENTS DOCUMENT

Re:Learn

Adaptive Multimodal Learning Environment

An AI tutor that finds the misconception behind a wrong answer, fixes it with a targeted intervention, and proves the fix stuck.

Domain: introductory physics (Newtonian mechanics).

Version	1.0
Date	3 October 2026
Status	Prototype built; model training in progress
Problem statement	#3, Re:Learn

SECTION 1

Summary

Most learning platforms grade an answer as right or wrong and then show the solution. That hides the most useful information a wrong answer carries: **which mistaken idea produced it**. Two students can pick the same wrong option for completely different reasons, and they need completely different help.

Re:Learn is an AI-powered practice environment for introductory mechanics. A learner answers a question and explains their reasoning in a sentence. A trained model reads the answer and the explanation together, names the underlying misconception, and the system delivers an intervention built for that misconception, including a short physics animation. The learner is then tested on a **new situation that exercises the same idea**. The misconception counts as resolved only if both the answer and the reasoning hold up.

One-line pitch: Re:Learn doesn't tell you that you're wrong. It tells you *why*, fixes that, and checks that the fix stuck.

Example: why right/wrong grading is not enough

Student	Picked	Explanation	Real misconception
A	Acceleration = 0	"It stops at the top, so velocity is zero."	Velocity–acceleration confusion
R	Acceleration = 0	"The force of the throw has run out."	Impetus (motion carries a force)
S	9.8 m/s ² down (correct)	"It is still slowing down because of the throw force."	Impetus, hidden behind a correct answer

A traditional LMS marks A and R identically and marks S as fully correct. Re:Learn separates all three.

SECTION 2

Problem

- **Fixed sequences.** LMS platforms push every learner through the same lectures, videos and quizzes.
- **Shallow adaptivity.** Adaptive systems mostly change question *difficulty*, not *how a concept is taught*.
- **Wrong = wrong.** An incorrect answer is treated as a binary signal, even though it often points to a specific, persistent misconception.
- **Showing the answer doesn't fix the idea.** A learner can memorise the right answer while keeping the wrong mental model.
- **Correct follow-ups are misleading.** Getting the next question right does not prove learning happened. The learner may have guessed, pattern-matched or memorised it.

SECTION 3

Goals and non-goals

Goals

- **G1. Diagnose:** infer the misconception behind a response from the chosen answer plus free-text reasoning.
- **G2. Differentiate:** tell apart misconceptions that produce the same wrong answer.

- **G3. Intervene:** deliver an explanation, worked example and visual simulation targeted at the diagnosed misconception.
- **G4. Verify:** reassess with transfer questions and judge resolution on answer *and* reasoning.
- **G5. Remember:** keep a per-learner model of active, recurring and resolved misconceptions.
- **G6. Prove it works:** publish diagnosis accuracy, including on questions and misconceptions not seen in training.

Non-goals (this version)

- Domains beyond mechanics (algebra and programming are future extensions).
- Full course content, video lectures or a teacher-authoring CMS.
- Account system, payments or multi-tenant school deployment.
- Handwriting/image input. Text explanations only for v1.

SECTION 4

Users

Persona	Who	Needs	What Re:Learn gives them
Learner	High-school / first-year physics student	Understand <i>why</i> they keep getting a type of question wrong	A named misconception, a targeted fix and proof it's gone
Teacher	Physics teacher or TA	Know which misconceptions a class holds, not just scores	Per-learner misconception map (class view planned)
Researcher / judge	Evaluator of the system	Evidence the model diagnoses correctly and generalises	Held-out evaluation, confusion matrix, unseen-misconception tests

SECTION 5

User journey

Step	What the learner does	What the system does
1. Start	Enters name	Loads questions and the learner's thinking map
2. Answer	Chooses an option, sets confidence (guessing / fairly sure / certain), writes one or two sentences of reasoning	Captures answer, explanation and confidence
3. Diagnose	Sees the verdict	Model returns one of: correct , flawed reasoning (right answer, wrong idea), misconception , or unknown (asks the learner to explain more)
4. Intervene	Reads the fix and watches the simulation	Shows the misconception name and description, model confidence, targeted explanation, worked example and animation
5. Reassess	Answers a transfer question in a new context, with reasoning	Re-diagnoses. Resolved only if the answer is right and the reasoning no longer shows the targeted misconception
6. Track	Sees the thinking map update	Marks each misconception active, recurring or resolved
7. Report	Finishes the session	Summary: right first time, found, fixed, still open

SECTION 6

Functional requirements

Priority: **P0** = required for the problem statement, **P1** = strongly expected, **P2** = nice to have.

ID	Requirement	Pri.	Status
FR-1	Misconception dataset: labelled (question, answer, explanation, misconception) rows covering correct, incorrect and right-answer-wrong-reason responses	P0	Planned
FR-2	Misconception model: trained classifier returning label + confidence. Replaces the keyword placeholder	P0	Partial
FR-3	Differentiation: questions map each wrong option to all misconceptions that can produce it; the model uses the explanation to choose between them	P0	Built
FR-4	Flawed-reasoning detection: flag misconceptions hidden behind correct answers	P0	Built
FR-5	Adaptive intervention: explanation, worked example and simulation per misconception	P0	Built
FR-6	Resolution assessment: transfer question; resolved = correct answer AND target misconception absent	P0	Built
FR-7	Stronger resolution: 2–3 transfer probes in different contexts plus a delayed recheck later in the session	P1	Planned
FR-8	Learner model: per-learner history with active / recurring / resolved states	P0	Built

ID	Requirement	Pri.	Status
FR-9	Unknown handling: if reasoning is insufficient, ask the learner to elaborate instead of guessing	P1	Built
FR-10	Unseen-misconception detection: low-confidence or embedding-similarity fallback against misconception descriptions so new misconceptions work without retraining	P1	Planned
FR-11	Evaluation suite: accuracy, macro-F1, confusion matrix, held-out questions, held-out misconceptions	P0	Planned
FR-12	Confidence capture: self-rated certainty, used to flag confidently held misconceptions	P2	Built
FR-13	Session report and progress indicators	P2	Built
FR-14	Teacher class view: misconception frequency across learners	P2	Planned

SECTION 7

Misconception taxonomy (v1)

Five well-documented mechanics misconceptions, drawn from physics-education research (e.g. Force Concept Inventory). Each needs an entry in the misconception list, the interventions and the reassessment bank.

ID	Name	Learner believes	Typical confusable with
VA_CONFUSION	Velocity–acceleration confusion	Zero velocity means zero acceleration	IMPETUS
IMPETUS	Impetus: motion carries a force	A moving object carries a force that keeps it going until it runs out	VA_CONFUSION, FORCE_VELOCITY
FORCE_VELOCITY	Force causes constant speed	A constant net force gives a constant speed	IMPETUS
THIRD_LAW	Unequal action–reaction	The heavier or faster object exerts the bigger force	HEAVIER_FASTER
HEAVIER_FASTER	Heavier objects fall faster	Heavier objects accelerate faster in free fall	THIRD_LAW

SECTION 8

Model and data

Input and output

Input: Q: {question} | **Answer:** {chosen option text} | **Explanation:** {reasoning}. **Output:** a misconception label (or NONE) with a confidence score. The service combines this with answer correctness to produce one of four statuses: correct, flawed_reasoning, misconception, unknown.

Dataset plan

- **Questions:** 25–40 multiple-choice items across kinematics, Newton's three laws and free fall. Each wrong option is mapped to the misconceptions that can produce it.
- **Responses:** target 1,500–2,500 labelled rows. Seed with hand-written explanations, expand with LLM-generated student-style paraphrases per (question, option, misconception), and include NONE rows with correct reasoning.
- **Hard cases on purpose:** same answer with different misconceptions, correct answers with flawed reasoning, short or vague explanations.
- **Quality:** manually review a random 10–15% sample. Report the review agreement rate.
- **Split by question**, never by row, so test questions are fully unseen.

Model

- **Classifier:** fine-tuned **DeBERTa-v3-base** sequence classifier (training script already in the repo).
- **Answer prior:** at inference, restrict or up-weight labels to those listed in the chosen option's misconception map. This cuts confusion between unrelated labels.
- **Unseen misconceptions:** a sentence-embedding model scores the explanation against every misconception description. When the classifier's confidence is low and similarity to an unseen description is high, use that label. New misconceptions can then be added by writing a description, with no retraining.
- **Abstain:** below a confidence threshold, return *unknown* and ask the learner to explain more.

SECTION 9

Resolution assessment

A correct follow-up answer is not treated as proof of learning. A misconception is **resolved** only when:

- the learner answers a **transfer question** correctly. It tests the same idea in a different context (e.g. a thrown ball becomes a pendulum, a truck becomes a mosquito);
- **and** the model finds no trace of the targeted misconception in the learner's explanation;
- **(planned)** and this holds across 2–3 probes and a delayed recheck later in the session.

If the same misconception is detected again after being resolved, it is re-opened and marked **recurring**.

SECTION 10

Success metrics

Metric	Definition	Target
Diagnosis macro-F1	On held-out questions	≥ 0.80
Confusable-pair accuracy	Accuracy on rows where the chosen option maps to 2+ misconceptions	≥ 0.75
Flawed-reasoning recall	Share of right-answer-wrong-reason rows correctly flagged	≥ 0.70
Unseen-misconception top-1	Accuracy on misconceptions held out of training (embedding fallback)	≥ 0.60
Calibration	Expected calibration error of confidence scores	≤ 0.10
Resolution rate	Share of diagnosed misconceptions resolved after one intervention	Report (no target in v1)
Learner effort	Share of answers that include an explanation	$\geq 80\%$

SECTION 11

System design

Component	Technology	Responsibility
Frontend	Single-page HTML/CSS/JS + canvas	Question flow, diagnosis, simulations, thinking map, report
API	FastAPI (Python)	Serves questions, runs diagnosis, records attempts, computes the learner profile
Model	Hugging Face Transformers (DeBERTa-v3)	diagnose(question, choice, explanation) returns status, label, confidence
Content	data.py	Misconceptions, questions + option maps, interventions, transfer questions
Storage	SQLite	Every attempt: learner, phase, choice, explanation, label, confidence, resolved

API

Endpoint	Purpose
GET /api/questions	Question list without answer keys
POST /api/diagnose	Diagnose a response. Returns status, misconception, intervention and transfer question
POST /api/reassess	Diagnose a transfer response against a target misconception. Returns resolved true/false
GET /api/profile/{learner}	Learner model: answered count and per-misconception detected / resolved / status

SECTION 12

Non-functional requirements

- **Latency:** diagnosis under 1 s on a laptop CPU.
- **Privacy:** answer keys are never sent to the browser. Learner data stays in a local database.
- **Accessibility:** full keyboard use, visible focus states, reduced-motion support, works at phone width.
- **Transparency:** always show model confidence. Abstain rather than guess on weak evidence.
- **Extensibility:** a new misconception only needs a description, an intervention and a transfer question.

SECTION 13

Milestones

#	Milestone	Deliverable	Status
M1	Prototype	End-to-end flow with rule-based diagnosis, learner model, redesigned UI	Built
M2	Dataset	Question bank expanded + 1,500+ labelled responses, split by question	Planned
M3	Model	Fine-tuned DeBERTa + answer prior, swapped in for the placeholder	Partial
M4	Evaluation	Metrics script, confusion matrix, held-out question and misconception results	Planned
M5	Generalisation	Embedding fallback for unseen misconceptions + abstain threshold	Planned
M6	Stronger reassessment	Multi-probe transfer + delayed recheck + recurring logic	Planned
M7	Demo	Scripted demo covering all four statuses and a resolved / not-resolved case	Planned

SECTION 14

Risks and mitigations

Risk	Impact	Mitigation
Synthetic data doesn't look like real student writing	Model overfits to LLM phrasing	Hand-written seed set, varied prompts, manual review, report results on hand-written rows separately
Short or empty explanations	Not enough signal to diagnose	Unknown status prompts for more detail. Sentence starters in the UI
Confusable misconceptions	Wrong intervention shown	Answer prior, targeted hard-case data, confusion-matrix tracking
Correct follow-up mistaken for learning	False 'resolved'	Reasoning must be clean too. Multiple transfer probes and delayed recheck
Overconfident model	Learners trust wrong diagnoses	Calibration check, displayed confidence, abstain threshold

SECTION 15

Open questions

- Should a learner be able to say "that's not what I meant" on a diagnosis, and should that feed back into training data?
- How long should the delay be before a resolved misconception is rechecked: same session, or next session?
- Is free-text the right input for every learner, or should there be a structured reasoning picker as a fallback?
- Which second domain (algebra or programming) best shows the approach generalises?